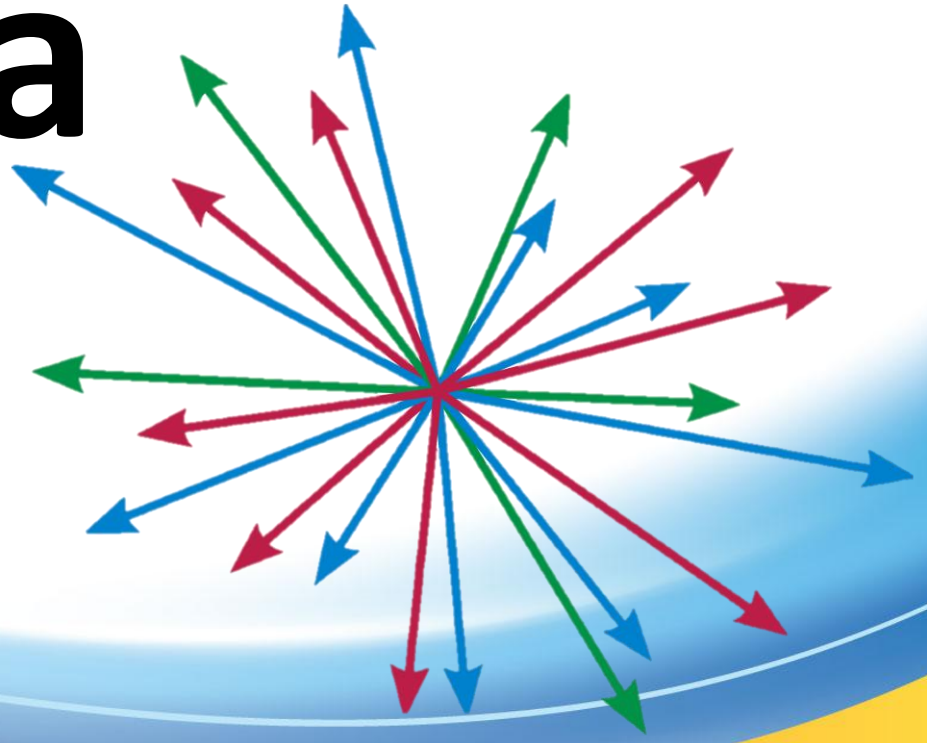
Undergraduate Course in Mathematics

Linear Algebra



Topic: Gram–Schmidt Procedure

Conducted By

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Orthogonal Complement

What is Orthogonal Complement

Let S be a subset of a vector space V and and $\langle \vec{\ }, \vec{\ } \rangle$ be an inner product. The orthogonal complement of S is denoted by $S^{\perp}$ and defined by

$$\langle \vec{u}, \vec{v} \rangle = 0 \text{ for all } \vec{u} \in S \text{ and } \vec{v} \in S^{\perp}$$

Calculate the orthogonal complement, $W^{\perp}$, (*i.e.* all the vectors in $W^{\perp}$ is orthogonal to all the vectors in $\boldsymbol{W}$) of the subspace

$$\boldsymbol{W} = \boldsymbol{span}\left\{\begin{pmatrix}\mathbf{1}\\ \mathbf{2}\\ \mathbf{-1}\end{pmatrix}, \begin{pmatrix}\mathbf{2}\\ \mathbf{4}\\ \mathbf{1}\end{pmatrix}\right\}$$

let $\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} \in W^{\perp}$.

$$\left.\begin{aligned} \therefore \left\langle \begin{pmatrix}1\\ 2\\ -1\end{pmatrix}, \begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} \right\rangle = 0 &\Rightarrow x_1 + 2x_2 - x_3 = 0 \\ \therefore \left\langle \begin{pmatrix}2\\ 4\\ 1\end{pmatrix}, \begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} \right\rangle = 0 &\Rightarrow 2x_1 + 4x_2 + x_3 = 0 \end{aligned}\right\}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 4 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right) \quad R_2' = R_2 - 2R_1$$

$$\left(\begin{array}{ccc|c} 3 & 6 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right) \quad R_1' = 3R_1 + R_2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\Rightarrow \begin{array}{l} x_1 + 2x_2 = 0 \\ x_3 = 0 \end{array}$$

free $x_2 = t$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2t \\ t \\ 0 \end{pmatrix} = t \cdot \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$\therefore W^{\perp} = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\}$$

Calculate the orthogonal complement, $W^{\perp}$, (*i.e.* all the vectors in $W^{\perp}$ is orthogonal to all the vectors in $\boldsymbol{W}$) of the subspace

$$\boldsymbol{W} = \boldsymbol{span}\left\{\begin{pmatrix}\mathbf{1}\\ \mathbf{1}\\ \mathbf{-1}\\ \mathbf{3}\end{pmatrix}, \begin{pmatrix}\mathbf{2}\\ \mathbf{1}\\ \mathbf{-2}\\ \mathbf{1}\end{pmatrix}\right\}$$

let $\begin{pmatrix}x_1\\ x_2\\ x_3\\ x_4\end{pmatrix} \in W^{\perp}$; $\therefore \left\langle \begin{pmatrix}1\\ 1\\ -1\\ 3\end{pmatrix}, \begin{pmatrix}x_1\\ x_2\\ x_3\\ x_4\end{pmatrix} \right\rangle = 0 \Rightarrow x_1 + x_2 - x_3 + 3x_4 = 0$

$\therefore \left\langle \begin{pmatrix}2\\ 1\\ -2\\ 1\end{pmatrix}, \begin{pmatrix}x_1\\ x_2\\ x_3\\ x_4\end{pmatrix} \right\rangle = 0 \Rightarrow 2x_1 + x_2 - 2x_3 + x_4 = 0$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ 2 & 1 & -2 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ 0 & -1 & 0 & -5 & 0 \end{array}\right) \quad R_2' = R_2 - 2R_1$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & -1 & 0 & -5 & 0 \end{array}\right) \quad R_1' = R_1 + R_2$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 0 & 5 & 0 \end{array}\right)$$

$$x_1 - x_3 - 2x_4 = 0$$
$$x_2 + 5x_4 = 0$$

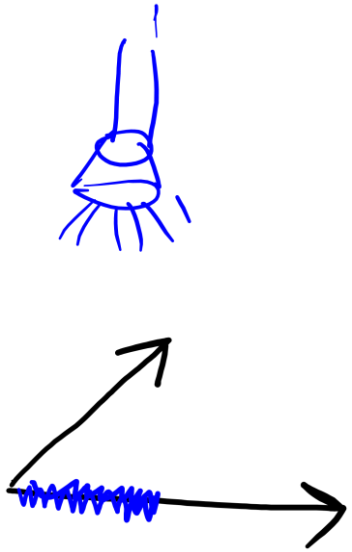
Let $x_3 = t_1$, $x_4 = t_2$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} t_1 + 2t_2 \\ -5t_2 \\ t_1 \\ t_2 \end{pmatrix}$$

$$= t_1 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -5 \\ 0 \\ 1 \end{pmatrix}$$

$$W^{\perp} = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -5 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Orthogonal Projection



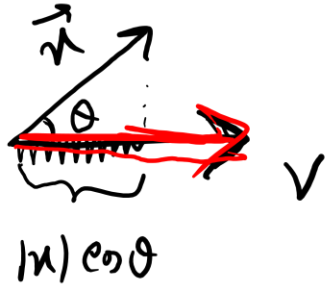
Orthogonal Projection of $\vec{x}$ onto $\vec{v}$

Let V be a subset of a vector space over F and and $\langle \overrightarrow{\quad}, \overrightarrow{\quad} \rangle$ be an inner product. The orthogonal projection of the vector $\vec{x}$ onto $\vec{v}$ is defined by

$$Proj_{\vec{v}}(\vec{x}) = \frac{\langle \vec{x}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$$

$$\vec{x} \cdot \vec{v} = |\vec{x}| \cdot |\vec{v}| \cdot \cos\theta$$

$$= |\vec{v}| \cdot \underline{|\vec{x}| \cos\theta} = |\vec{v}| \cdot \text{Projection}$$



$$\Rightarrow \text{Project} = \frac{\vec{x} \cdot \vec{v}}{|\vec{v}|}$$

with direction

$$\frac{x \cdot \vec{v}}{\|\vec{v}\|} \cdot \frac{\vec{v}}{\|\vec{v}\|} = \frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Let $\boldsymbol{u} = (8, \sqrt{3}, \sqrt{7}, -1, 1)$ and $\boldsymbol{v} = (1, -1, 0, 2, \sqrt{3})$. If the orthogonal projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$ is $\frac{a}{b}\boldsymbol{v}$, then determine $\boldsymbol{a}$ and $\boldsymbol{b}$.

$$\text{Proj}_{\vec{v}}(\vec{u}) = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v} = \frac{8 - \sqrt{3} + 0 - 2 + \sqrt{3}}{1^2 + (-1)^2 + 0^2 + 2^2 + (\sqrt{3})^2} \vec{v}$$

$$= \frac{6}{9} \vec{v}$$

$\therefore a = 6 \qquad b = 9$

Gram–Schmidt Procedure

Basis ⇒ Orthogonal Basis

Gram–Schmidt Procedure

Given $S = \{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}, \dots, \overrightarrow{v_n}\}$ a linearly independent set. The Gram–Schmidt Procedure constructs an Orthogonal Set $S' = \{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, \dots, \overrightarrow{w_n}\}$ such that

$$span\,\{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}, \dots, \overrightarrow{v_n}\} = span\,\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, \dots, \overrightarrow{w_n}\}$$

Furthermore, we can obtain an orthonormal set $S'' = \{\overrightarrow{e_1}, \overrightarrow{e_2}, \overrightarrow{e_3}, \dots, \overrightarrow{e_n}\}$ by dividing each vector by its own norm.

Orthogonal Vectors

$$\overrightarrow{w_1} = \overrightarrow{v_1}$$

$$\overrightarrow{w_2} = \overrightarrow{v_2} - \frac{\langle \overrightarrow{v_2}, \overrightarrow{w_1} \rangle}{\langle \overrightarrow{w_1}, \overrightarrow{w_1} \rangle} \overrightarrow{w_1}$$

$$\overrightarrow{w_3} = \overrightarrow{v_3} - \frac{\langle \overrightarrow{v_3}, \overrightarrow{w_1} \rangle}{\langle \overrightarrow{w_1}, \overrightarrow{w_1} \rangle} \overrightarrow{w_1} - \frac{\langle \overrightarrow{v_3}, \overrightarrow{w_2} \rangle}{\langle \overrightarrow{w_2}, \overrightarrow{w_2} \rangle} \overrightarrow{w_2}$$

$$\overrightarrow{w_4} = \overrightarrow{v_4} - \frac{\langle \overrightarrow{v_4}, \overrightarrow{w_1} \rangle}{\langle \overrightarrow{w_1}, \overrightarrow{w_1} \rangle} \overrightarrow{w_1} - \frac{\langle \overrightarrow{v_4}, \overrightarrow{w_2} \rangle}{\langle \overrightarrow{w_2}, \overrightarrow{w_2} \rangle} \overrightarrow{w_2} - \frac{\langle \overrightarrow{v_4}, \overrightarrow{w_3} \rangle}{\langle \overrightarrow{w_3}, \overrightarrow{w_3} \rangle} \overrightarrow{w_3}$$

$$\dots$$

$$\overrightarrow{w_n} = \overrightarrow{v_n} - \frac{\langle \overrightarrow{v_n}, \overrightarrow{w_1} \rangle}{\langle \overrightarrow{w_1}, \overrightarrow{w_1} \rangle} \overrightarrow{w_1} - \frac{\langle \overrightarrow{v_n}, \overrightarrow{w_2} \rangle}{\langle \overrightarrow{w_2}, \overrightarrow{w_2} \rangle} \overrightarrow{w_2} - \dots - \frac{\langle \overrightarrow{v_n}, \overrightarrow{w_{n-1}} \rangle}{\langle \overrightarrow{w_{n-1}}, \overrightarrow{w_{n-1}} \rangle} \overrightarrow{w_{n-1}}$$

Orthonormal

$$\overrightarrow{e_1} = \frac{\overrightarrow{w_1}}{\|\overrightarrow{w_1}\|}$$

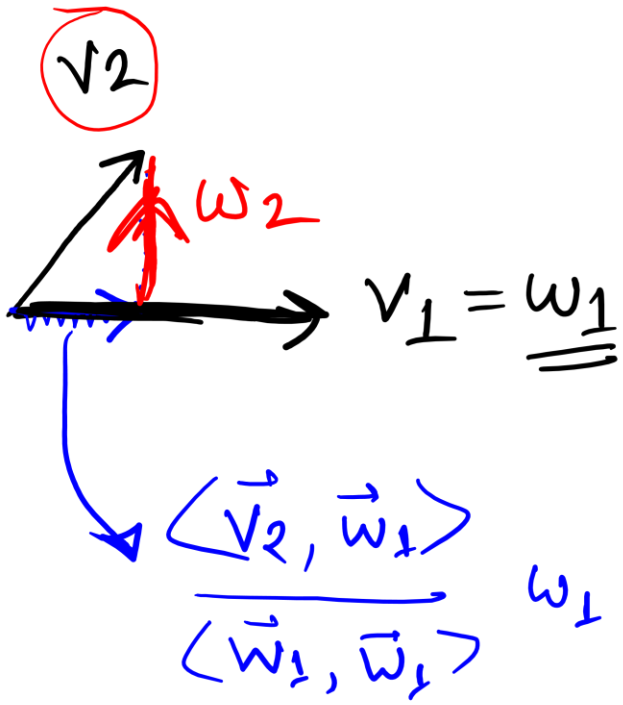
$$\overrightarrow{e_2} = \frac{\overrightarrow{w_1}}{\|\overrightarrow{w_2}\|}$$

$$\overrightarrow{e_3} = \frac{\overrightarrow{w_3}}{\|\overrightarrow{w_3}\|}$$

$$\overrightarrow{e_4} = \frac{\overrightarrow{w_4}}{\|\overrightarrow{w_4}\|}$$

$$\dots$$

$$\overrightarrow{e_n} = \frac{\overrightarrow{w_n}}{\|\overrightarrow{w_n}\|}$$



w_1

w_2

$$\text{red} = \underbrace{v_2 - \frac{\langle \vec{v}_2, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} w_1}_{w_2}$$

Use the Gram-Schmidt process to transform the basis $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ into an orthonormal basis.

$$\boldsymbol{v_1} = (\mathbf{1}, \mathbf{1}, \mathbf{1}), \qquad \boldsymbol{v_2} = (\mathbf{0}, \mathbf{1}, \mathbf{1}), \qquad \boldsymbol{v_3} = (\mathbf{0}, \mathbf{0}, \mathbf{1})$$

$$w_1 = (1,1,1)$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$= (0,1,1) - \frac{0+1+1}{1+1+1}(1,1,1) = (0,1,1) - \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$= \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$v_1 \quad (1,1,1) \qquad v_2 = (0,1,1) \qquad v_3 = (0,0,1)$

$w_1 = (1,1,1) \qquad w_2 = \left(\frac{-2}{3}, \frac{1}{3}, \frac{1}{3}\right)$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$= (0,0,1) - \frac{1}{3}(1,1,1) - \frac{\frac{1}{3}}{\frac{4}{9}+\frac{1}{9}+\frac{1}{9}}\left(\frac{-2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$= (0,0,1) - \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) - \frac{1/3}{6/9}\left(\frac{-2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$= (0,0,1) - \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) - \left(\frac{-1}{3}, \frac{1}{6}, \frac{1}{6}\right)$$

$$= \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$\vec{e}_1 = \frac{w_1}{\|w_1\|} = \frac{(1,1,1)}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$\vec{e}_2 = \frac{w_2}{\|w_2\|} = \frac{\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)}{\sqrt{\frac{4}{9}+\frac{1}{9}+\frac{1}{9}}} = \frac{\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)}{\frac{\sqrt{6}}{3}} = \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$\vec{e}_3 = \frac{w_3}{\|w_3\|} = \frac{\left(0, -\frac{1}{2}, \frac{1}{2}\right)}{\sqrt{\frac{1}{4}+\frac{1}{4}}} = \frac{\left(0, -\frac{1}{2}, \frac{1}{2}\right)}{\frac{1}{\sqrt{2}}} = \left(0, \frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

Use the Gram-Schmidt process to transform the basis $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ into an orthonormal basis.

$$\boldsymbol{v_1} = (\mathbf{1}, \mathbf{1}, \mathbf{0}), \qquad \boldsymbol{v_2} = (\mathbf{1}, \mathbf{0}, -\mathbf{2}), \qquad \boldsymbol{v_3} = (\mathbf{0}, \mathbf{4}, \mathbf{1})$$

$$w_1 = (1, 1, 0)$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1, 0, -2) - \frac{1+0+0}{1+1+0}(1, 1, 0)$$

$$= (1, 0, -2) - \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$= \left(\frac{1}{2}, \frac{-1}{2}, -2\right).$$

$$v_1 = (1,1,0), \qquad v_2 = (1,0,-2), \qquad v_3 = (0,4,1)$$

$$w_1 = (1,1,0) \qquad w_2 = \left(\frac{1}{2}, \frac{-1}{2}, -2\right)$$

$$\frac{\frac{-4}{1+1+16}}{4} = \frac{-16}{18} = \frac{-8}{9}$$

$$w_3 = v_3 - \frac{\langle v_3, w_1\rangle}{\langle w_1, w_1\rangle} w_1 - \frac{\langle v_3, w_2\rangle}{\langle w_2, w_2\rangle} w_2$$

$$= (0,4,1) - \frac{4}{2}(1,1,0) - \frac{0-2-2}{\frac{1}{4}+\frac{1}{4}+4}\left(\frac{1}{2}, \frac{-1}{2}, -2\right)$$

$$= (0,4,1) - (2,2,0) - \frac{-8}{9}\left(\frac{1}{2}, \frac{-1}{2}, -2\right)$$

$$(0,4,1) - (2,2,0) + \frac{+8}{9}\left(\frac{1}{2}, \frac{-1}{2}, -2\right)$$

$$= (0,4,1) - (2,2,0) + \left(\frac{4}{9}, -\frac{4}{9}, \frac{-16}{9}\right)$$

$$= \left(0-2+\frac{4}{9},\ 4-2-\frac{4}{9},\ 1-\frac{16}{9}\right)$$

$$= (\quad , \quad , \quad).$$

$$\vec{e}_1 = \frac{\vec{w}_1}{\|\vec{w}_1\|}$$

$\vec{e}_2$

$\vec{e}_3$

Advantages of Orthogonal Basis

Finding Linear Combination from Orthogonal Basis

$$\vec{V} = \underline{\underline{\alpha_1}}\,\vec{e}_1 + \underline{\underline{\alpha_2}}\,\vec{e}_2 + \underline{\underline{\alpha_3}}\,\vec{e}_3 + \cdots$$

Consider the orthogonal basis $\mathbf{B} = \{(\mathbf{1},\mathbf{1},\mathbf{1}), (1,-1,0), (1,1,-2)\}$. Express $(\mathbf{3},\mathbf{9},\mathbf{0})$ as a linear combination of the basis vectors.

$v_1 \quad v_2 \quad v_3$

$$\vec{v} = \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \alpha_3 \vec{v}_3$$

$$\langle \vec{v}, v_1 \rangle = \alpha_1 \langle \vec{v}_1, \vec{v}_1 \rangle + \alpha_2 \cancel{\langle \vec{v}_2, \vec{v}_1 \rangle} + \alpha_3 \cancel{\langle \vec{v}_3, \vec{v}_1 \rangle}$$

$$\Rightarrow \alpha_1 = \frac{\langle \vec{v}, v_1 \rangle}{\langle v_1, v_1 \rangle} \qquad \alpha_2 = \frac{\langle \vec{v}, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle} \qquad \alpha_3 = \frac{\langle \vec{v}, \vec{v}_3 \rangle}{\langle \vec{v}_3, \vec{v}_3 \rangle}$$

$$\alpha_1 = \frac{\langle (3,9,0), (1,1,1) \rangle}{\langle (1,1,1), (1,1,1) \rangle} = \frac{12}{3} = 4$$

$$\alpha_2 = \frac{\langle (3,9,0), (1,-1,0) \rangle}{\langle (1,-1,0), (1,-1,0) \rangle} = \frac{-6}{2} = -3$$

$$\alpha_3 = \frac{\langle (3,9,0), (1,1,-2) \rangle}{\langle (1,1,-2), (1,1,-2) \rangle} = \frac{12}{6} = 2.$$

$$(3,9,0) = 4\cdot(1,1,1) - 3\cdot(1,-1,0) + 2\cdot(1,1,-2)$$

Fourier Series

Vector Space of periodic functions

$V = \{$ all periodic function of period $2\pi \}$

is a vector space

Fourier Basis is Orthogonal

$$B = \{1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \sin 3x, \ldots\}$$

orthogonal Basis

$$\langle f(x), g(x) \rangle = \int_{-\pi}^{\pi} f(x) \cdot g(x)\, dx$$

Fourier Series

$$f(x) = a_0 \cdot 1 + a_1 \cdot \cos x + a_2 \cos 2x + a_3 \cdot \cos 3x + \cdots$$
$$+ b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \cdots$$

$$a_2 = \frac{\langle f(x), \cos 2x \rangle}{\langle \cos 2x, \cos 2x \rangle} = \frac{\int_{-\pi}^{\pi} f(x) \cos 2x \, dx}{\pi}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx$$

$(-\pi, \pi)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx$$

General Fourier Series

$(-L, L)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx \qquad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Thank you!